

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276829

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Vinger; Florence M.

RECLOSABLE CAN ENDS

Abstract

Reclosable can ends are provided. In some embodiments, the reclosable can ends comprise: a lid base that comprises a scored area; a tab that causes the scored area to at least partially separate from the lid base to expose an opening in the lid base; a closure flap, wherein the closure flap comprises: a first portion; a second portion; an opening; and a protruding portion; and a rivet that connects the tab and the closure flap to the lid base such that the tab and the closure flap are configured to rotate around the rivet, wherein the first portion of the closure flap is configured to be inserted into the opening in the lid base end when the tab and the closure flap are rotated to a position corresponding to a position of the opening in the lid base.

Inventors: Vinger; Florence M. (Los Angeles, CA)

Applicant: Vinger; Florence M. (Los Angeles, CA)

Family ID: 96880897

Appl. No.: 18/141677

Filed: May 01, 2023

Related U.S. Application Data

parent US continuation-in-part 16853237 20200420 parent-grant-document US 11661233 child US 18141677

us-provisional-application US 63455956 20230330

us-provisional-application US 62868709 20190628

Publication Classification

Int. Cl.: B65D17/28 (20060101)

U.S. Cl.:

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/455,956, filed Mar. 30, 2023, and is a continuation-in-part of U.S. patent application Ser. No. 16/853,237, filed Apr. 20, 2020, which claims the benefit of U.S. Provisional Application No. 62/868,709, filed Jun. 28, 2019, each of which is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The disclosed subject matter relates to reclosable can ends.

BACKGROUND

[0003] Cans are frequently used for storing and drinking such beverages as soda, beer, water, Ready To Drink (RTD) beverages and cocktails, energy drinks, juices, sparkling water and many more beverages and liquids, and/or for other cannable storage needs. Cans, which are often made from aluminum and aluminum alloy, are convenient for storing and drinking for many reasons. For example, aluminum is lightweight, is easily transportable, saves space in shipping containers, is sturdy, and is 100% recyclable. However, once a can is opened, it is typically not used as a portable container. In particular, cans are typically not reclosable, and it can therefore be difficult to transport an opened can and to keep a beverage in an opened can free from dirt, debris, or bugs.

[0004] Accordingly, it is desirable to provide reclosable can ends.

SUMMARY

[0005] In accordance with some embodiments, reclosable can ends are provided.

[0006] In some embodiments, reclosable can ends are provided, the reclosable can ends comprising: a lid base, wherein the lid base comprises a scored area; a tab, wherein a first end of the tab is configured to apply pressure to the scored area of the lid base when a second end of the tab, opposite the first end of the tab, is lifted and causes the scored area to at least partially separate from the lid base to expose an opening in the lid base; a closure flap, wherein the closure flap comprises: a first portion that has dimensions corresponding to dimensions of the opening in the lid base, wherein the first portion of the closure flap connects with a portion of the tab; a second portion that comprises an outer edge that extends beyond the first portion of the closure flap; a protruding portion that protrudes from the outer edge of the second portion of the closure flap and includes an opening; at least one bump that can contact the second end of the tab when tab is in a downward position; and a rivet that connects the tab and the closure flap to the lid base such that the tab and the closure flap are configured to rotate around the rivet, wherein the rivet connects the tab and the closure flap to the lid base via the opening in the protruding portion of the closure flap, wherein the first portion of the closure flap is configured to be inserted into the opening in the lid base when the tab and the closure flap are rotated to a position corresponding to a position of the opening in the lid base, and wherein the outer edge of the second portion of the closure flap is positioned on a top portion of the lid base when the first portion of the closure flap is inserted into the opening in the lid base. In some of these embodiments, the lid base comprises a lid base pocket that has dimensions that correspond to the dimensions of the first portion of the closure flap. In some of these embodiments, the lid base comprises a depressed slide area formed from the lid base pocket to the opening in the lid base. In some of these embodiments, a bottom portion of the first portion of the closure flap includes a coating of sealant. In some of these embodiments, the coating of sealant comprises a plurality of layers of sealant. In some of these embodiments, the coating of sealant comprises a material that is harmless to consumers. In some of these embodiments, the second end of the tab is concave such that the second end curves toward the first end. In some of

these embodiments, the first portion of the closure flap includes a raised bump that connects the closure flap to the tab. In some of these embodiments, the outer edge of the second portion of the closure flap is flat. In some of these embodiments, the outer edge of the second portion of the closure flap is angled upward away from the first portion of the closure flap. In some of these embodiments, the outer edge of the second portion of the closure flap has an arched shape that bends away from the first portion of the closure flap. In some of these embodiments, the outer edge of the second portion of the closure flap has an arched shape that bends towards the first portion of the closure flap. In some of these embodiments, the outer edge of the second portion of the closure flap has an arched shape. In some of these embodiments, the rivet is formed as part of the top portion of the lid base. In some of these embodiments, the rivet is formed as part of the tab. In some of these embodiments, the rivet is not formed from any of the lid base, the tab, and the closure flap. In some of these embodiments, the lid base comprises a drip catch.

[0007] In some embodiments, reclosable can ends are provided, the reclosable can ends comprising: a lid base, wherein the lid base comprises a scored area; a tab, wherein a first end of the tab is configured to apply pressure to the scored area of the lid base when a second end of the tab, opposite the first end of the tab, is lifted and causes the scored area to at least partially separate from the lid base to expose an opening in the lid base; a closure flap, wherein the closure flap comprises: a first portion that has dimensions corresponding to dimensions of the opening in the lid base, wherein the first portion of the closure flap connects with a portion of the tab; a second portion that comprises an outer edge that extends beyond the first portion of the closure flap; and a protruding portion that protrudes from the outer edge of the second portion of the closure flap and includes an opening; and a rivet that connects the tab and the closure flap to the lid base such that the tab and the closure flap are configured to rotate around the rivet, wherein the rivet connects the tab and the closure flap to the lid base via the opening in the protruding portion of the closure flap, wherein the first portion of the closure flap is configured to be inserted into the opening in the lid base when the tab and the closure flap are rotated to a position corresponding to a position of the opening in the lid base, wherein the outer edge of the second portion of the closure flap is positioned on the top portion of the lid base when the first portion of the closure flap is inserted into the opening in the lid base, and wherein the lid base comprises a push button that is positioned to serve as a stop against movement of the closure flap and the tab, but may be selectively depressed to permit movement of the closure flap and the tab. In some of these embodiments, the lid base comprises a lid base pocket that has dimensions that correspond to the dimensions of the first portion of the closure flap. In some of these embodiments, the lid base comprises a depressed slide area formed from the lid base pocket to the opening in the lid base.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Various objects, features, and advantages of the disclosed subject matter can be more fully appreciated with reference to the following detailed description of the disclosed subject matter when considered in connection with the following drawings, in which like reference numerals identify like elements.

[0009] FIGS. 1A and 1B show top views of a reclosable can end with an attached lid base, tab, rivet, and closure flap in accordance with some embodiments of the disclosed subject matter.

[0010] FIG. 2A shows a top view of reclosable can end with a lid base and a closure flap in accordance with some embodiments of the disclosed subject matter.

[0011] FIG. 2B shows a top view of a reclosable can end with a lid base and without an attached tab, rivet, or closure flap in accordance with some embodiments of the disclosed subject matter.

[0012] FIGS. 3A, 3B, 3C, and 3D show views of a tab in accordance with some embodiments of

the disclosed subject matter.

[0013] FIG. 3E shows views of a tab that has a rivet portion in accordance with some embodiments of the disclosed subject matter.

[0014] FIG. 3F shows an underside of a lid base with an attached rivet in accordance with some embodiments of the disclosed subject matter.

[0015] FIGS. 4A, 4B, and 4C show example views of an independent rivet for attaching a tab, a closure flap, and a lid base in accordance with some embodiments of the disclosed subject matter.

[0016] FIGS. 5A, 5B, 5C, and 5D show example views of a scored flap and perforations corresponding to a can opening in accordance with some embodiments of the disclosed subject matter.

[0017] FIGS. 6A, 6B, and 6C, show views of a closure flap in accordance with some embodiments of the disclosed subject matter.

[0018] FIGS. 7A, 7B, and 7C show cross-sectional side views of a lid base, a closure flap, and a tab in accordance with some embodiments of the disclosed subject matter.

[0019] FIG. 8 shows example cross-sectional side views of shapes for closure flaps in accordance with some embodiments of the disclosed subject matter.

[0020] FIG. 9A shows a top view of a can end with an attached tab, closure flap, and rivet in accordance with some embodiments of the disclosed subject matter.

[0021] FIG. 9B shows an example view of a closure flap with a closure flap rivet and a tab in accordance with some embodiments of the disclosed subject matter.

[0022] FIGS. 9C and 9D show example views of a closure flap that can connect to a tab and/or a lid base rivet in accordance with some embodiments of the disclosed subject matter.

[0023] FIG. 9E shows an additional closure flap rivet to connect a pull tab to a closure rivet in accordance with some embodiments of the disclosed subject matter.

[0024] FIG. 9F shows an example view of a tab in accordance with some embodiments of the disclosed subject matter.

[0025] FIG. 9G shows a cross-sectional view of a flattened rivet in accordance with some embodiments of the disclosed subject matter.

[0026] FIGS. 10A, 10B, and 10C show example views of lid bases with different lid base rivet areas in accordance with some embodiments of the disclosed subject matter.

[0027] FIG. 11 shows an example of a lid base with a slide area for a closure flap and a push button in accordance with some embodiments of the disclosed subject matter.

[0028] FIG. 12 shows a top view of an example of a can end assembly in accordance with some embodiments of the disclosed subject matter.

[0029] FIG. 13 shows a perspective view of the example of a can end assembly of FIG. 12 in accordance with some embodiments of the disclosed subject matter.

[0030] FIG. 14 shows a cross-sectional view of the example of a can end assembly of FIG. 12 in accordance with some embodiments of the disclosed subject matter.

[0031] FIG. 15 shows a perspective view of an example of a can end in accordance with some embodiments of the disclosed subject matter.

[0032] FIG. 16 shows a top view of the example of a can end of FIG. 15 in accordance with some embodiments of the disclosed subject matter.

[0033] FIG. 17 shows a top view of an example of a closure on a can end in accordance with some embodiments of the disclosed subject matter.

[0034] FIG. 18 shows a top view of the example of a closure of FIG. 17 in accordance with some embodiments of the disclosed subject matter.

[0035] FIG. 19 shows a top view of another example of a closure in accordance with some embodiments of the disclosed subject matter.

[0036] FIG. 20 shows a perspective view of the example of a closure of FIG. 19 in accordance with some embodiments of the disclosed subject matter.

[0037] FIG. **21** shows a cross-section view of the example of a closure of FIG. **17** or FIG. **19** in accordance with some embodiments of the disclosed subject matter.

[0038] FIG. **22** shows a top view of an example of a tab in accordance with some embodiments of the disclosed subject matter.

[0039] FIG. **23** shows a perspective view of the example of a tab of FIG. **22** in accordance with some embodiments of the disclosed subject matter.

[0040] FIG. **24** shows a cross-sectional view of the example of a tab of FIG. **22** in accordance with some embodiments of the disclosed subject matter.

[0041] FIG. **25** shows a top view of an example of a can end assembly having a safety feature in accordance with some embodiments of the disclosed subject matter.

[0042] FIG. **26** shows a top view of an example of the closure of FIG. **25** in accordance with some embodiments of the disclosed subject matter.

[0043] FIG. **27** shows a perspective view of an example of a closure of FIG. **25** in accordance with some embodiments of the disclosed subject matter.

[0044] FIG. **28** shows a top view of an example of a can end assembly having another safety feature in accordance with some embodiments of the disclosed subject matter.

[0045] FIG. **29** shows a top view of an example of the closure of FIG. **28** in accordance with some embodiments of the disclosed subject matter.

[0046] FIG. **30** shows a perspective view of an example of a closure of FIG. **28** in accordance with some embodiments of the disclosed subject matter.

DETAILED DESCRIPTION

[0047] In accordance with various embodiments, reclosable can ends are provided.

[0048] In some embodiments, reclosable can ends are described herein. In some embodiments, a reclosable can end can be part of any suitable type of can, such as a soda can, that can be used to store any suitable type of liquid or beverage, or a can that can be used for any other suitable cannable storage needs.

[0049] In some embodiments, a can that includes a reclosable can end can be opened. For example, in some embodiments, a lid base of the can end (e.g., a flat top portion of the can end) can be opened such that a scored flap of the lid base is pressed inward to expose an opening in the lid base. In some embodiments, a scored flap can be pressed inward in any suitable manner, such as by using a tab attached to the lid base that can act as a lever to apply pressure to the scored portion, thereby breaking a perforated border of the scored flap and lid base.

[0050] In some embodiments, the tab can be coupled to a closure flap that has a size and dimensions corresponding to an opening in the lid base such that the closure flap can be turned and then pressed into the opening to close the opening in the lid base, thereby providing a seal in the opening in the lid base. In some embodiments, the closure flap can be pulled and/or rotated out of the opening to open the lid base. Continuing further, in some embodiments, the closure flap can be reinserted in the lid base to reclose the opening in the lid base. In some embodiments, the closure flap can be inserted and rotated from an opening of an opened can any suitable number of times while still maintaining a seal over the opening.

[0051] In some embodiments, a tab can be coupled to a closure flap and attached to a lid base of a can end in any suitable manner. For example, in some embodiments, a tab can be coupled to a closure flap and attached to a lid base via a rivet made from a raised center area of the lid base, as shown in and described below in connection with FIGS. **10A**, **10B**, and **10C**. As another example, in some embodiments, a tab can be coupled to a closure flap and attached to a lid base via a rivet made from a raised center area of the tab, such as shown in and described below in connection with FIG. **3E**. As yet another example, in some embodiments, a tab can be coupled to a closure flap and attached to a lid base via an independent rivet, such as shown in and described below in connection with FIGS. **4A**, **4B**, and **4C**. In some embodiments, any suitable mechanism can be used to couple a tab to a closure flap and to attach the coupled tab and closure flap to a lid base, which can allow

the tab and the closure flap to lock and rotate together.

[0052] Note that, in some embodiments, “can end” can refer to a top portion of a can. In some embodiments, a can end can include any suitable elements, such as a lid base (e.g., a flat portion of a can end with shaped edges to allow the can end to attach to a can body), a scored flap corresponding to an opening in the lid base when the can has been opened, a closure flap, a tab, and/or any suitable rivets. Additionally, note that, in some embodiments, lid bases, closure flaps, tabs, and rivet(s) as described herein can be manufactured for can ends of any suitable size.

[0053] In some embodiments, a closure flap, a tab, and any rivet(s) can remain attached to a can, thereby reducing litter and protecting consumers and animals from abrasions due to any sharp and/or metal edges. Additionally, in some embodiments, any suitable portions of a closure flap, a tab, rivet(s), a lid base, and/or any other suitable elements of a can end can be made from any suitable materials (e.g., aluminum and/or aluminum alloy as used in a can body), thereby allowing the closure flap, the tab, the rivet(s), and/or the lid base to be recycled.

[0054] Turning to FIGS. 1A and 1B, top views of examples of reclosable can ends are shown in accordance with some embodiments of the disclosed subject matter. FIG. 1A shows an example of a lid base **100** that has not yet been opened. Note that, in some embodiments, a can end can be made of any suitable material, such as aluminum, an aluminum alloy, steel, and/or any other suitable materials. Additionally, note that, in some embodiments, a body of a can can be made of a different material than a can end. In some embodiments, an internal portion of a can end can be coated with any suitable material or combination of materials. For example, in some embodiments, an internal portion of the can end can be coated with epoxy resin. As another example, in some embodiments, an internal portion of the can end can be coated with a polymer plastic lining that is made of any suitable material or combination of materials (e.g., polyethylene, polypropylene, polyester, styrene, acrylic, vinyl, and/or any other suitable material(s)).

[0055] As illustrated, FIG. 1A includes a tab **102**, a closure flap **108**, a tab rivet area **110**, lid base **100**, and a scored flap **104**. Note that, as shown in FIG. 1A, scored flap **104** is fully attached to lid base **100**, because tab **102** has not yet caused scored flap **104** to separate from a top portion of lid base **100**. In some embodiments, tab **102** can be connected to any suitable portion of lid base **100** (e.g., a center portion, and/or any other suitable portion) in any suitable manner. For example, in some embodiments, tab **102** can be connected to lid base **100** via a rivet in tab rivet area **110**. Note that, in some such embodiments, a rivet in tab rivet area **110** can be a rivet that is part of lid base **100** (e.g., as shown in and described below in connection with FIGS. 10A, 10B, and/or 10C), a rivet that is part of tab **102** in tab rivet area **110** (e.g., as shown in and described below in connection with FIG. 3E), and/or an independent rivet inserted in tab rivet area **110** (e.g., as shown in and described below in connection with FIGS. 4A, 4B, and 4C).

[0056] In some embodiments, tab **102** can cause a portion of scored flap **104** to at least partially separate from lid base **100**, thereby opening lid base **100**, as illustrated in FIG. 1B. In some embodiments, opening lid base **100** can cause an opening **106** of lid base **100** to be exposed. In some embodiments, tab **102** can cause the portion of scored flap **104** to separate from lid base **100** in any suitable manner. For example, in some embodiments, a tip of tab **102** that is above scored flap **104** can be pressed into scored flap **104**, thereby causing a portion of scored flap **104** to be pressed into an interior of a can body attached to lid base **100**. In some embodiments, the tip of tab **102** can be pressed into scored flap **104** in any suitable manner. For example, in some embodiments, tab **102** can be operated as a second class lever and a first class lever around a corresponding to tab rivet area **110** by lifting a back end of tab **102** that is opposite the tip of tab **102**, thereby causing the tip of tab **102** to apply pressure to scored flap **104**.

[0057] As shown in FIG. 1B, in some embodiments, tab **102** can be connected to a closure flap **108** in any suitable manner including a rivet, welding, recyclable sealant, rubber, recyclable glue, links, and/or in any other suitable manner. For example, in some embodiments, tab **102** can be connected to closure flap **108** via a rivet corresponding to tab rivet area **110**. Note that, in some embodiments,

the rivet corresponding to tab rivet area **110** can be part of lid base **100** (e.g., as shown in and described below in connection with FIG. **10A**), part of tab **102** (e.g., as shown in and described below in connection with FIG. **3E**), and/or an independent rivet (e.g., as shown in and described below in connection with FIGS. **4A**, **4B**, and **4C**).

[0058] As shown in FIG. **1B**, in some embodiments, closure flap **108** can have a size and a shape that corresponds to opening **106**, thereby allowing closure flap **108** to be inserted into opening **106** to re-close and seal lid base **100** after scored flap **104** of lid base **100** has been opened.

[0059] Note that, as shown in FIG. **1B**, tab **102** can be joined by a rivet corresponding to tab rivet area **110**, thereby allowing tab **102** to rotate in conjunction with closure flap **108**, either clockwise or counterclockwise. For example, in some embodiments, closure flap **108** can be rotated to a position suitable for insertion of closure flap **108** into opening **106** to seal opening **106**.

[0060] Additionally, note that, in some embodiments, scored flap **104** can have any suitable size and/or any suitable shape depressions. In some embodiments, scored flap **104** can be scored or perforated to allow a portion of scored flap **104** to separate from lid base **100** when a tip of tab **102** is depressed on scored flap **104** to open lid base **100**. Note that, when opened, in some embodiments, scored flap **104** can remain attached to lid base **100** via a link **112**. In some embodiments, any suitable portion of scored flap **104**, link **112**, and/or lid base **100** can be scored or perforated to allow any portion of scored flap **104** to bend from lid base **100** when lid base **100** is opened to allow the portion of scored flap **104** to remain attached to lid base **100** while providing a seal between closure flap **108** and opening **106**.

[0061] In some embodiments, scored flap **104** can be scored in multiple ways around a circumference of opening **106** and at link **112**. In some embodiments, scored flap **104**, where it meets link **112**, can curl upward and inward toward the center of lid base **100**, such as shown in and described below in connection with FIG. **5C**. As another example, in some embodiments, scored flap **104**, where it meets link **112**, can be scored such that it is a dead end stop with no curl as shown in and described below in connection with FIG. **5A**. As yet another example, in some embodiments, scored flap **104**, where it meets link **112**, can be scored such that it can curl down and inward toward the center of scored flap **104** as shown in and described below in connection with FIG. **5D**.

[0062] In some embodiments, an underside of link **112** can be perforated in any suitable manner (e.g., as shown in and described below in connection with FIG. **5B**) to create a more desirable angle to provide a seal of closure flap **108** over opening **106** of lid base **100** and link **112**. For example, in some embodiments, the underside of link **112** adjacent to a point at which link **112** connects to lid base **100** when lid base **100** is open can have small cuts or indents such that scored flap **104** bends around the perforations of link **112** when lid base **100** is open. In some embodiments, a depth of the perforations of the underside of link **112** can be deep enough to allow scored flap **104** to bend straight inward into opening **106** and around link **112** area but shallow enough that link **112** remains connected to lid base **100** when lid base **100** is opened.

[0063] Turning to FIG. **2A**, an example of lid base **100** without a tab and when lid base **100** has not yet been opened (that is, scored flap **104** has not been pressed down) is shown in accordance with some embodiments of the disclosed subject matter. Note that, as shown in FIG. **2A**, closure flap **108** can be positioned in a portion (e.g., a recessed portion, and/or any other suitable portion) of lid base **100** when closure flap **108** is not in use.

[0064] Turning to FIG. **2B**, an example of lid base **100** without a tab and without a closure flap, when lid base **100** has not yet been opened is shown in accordance with some embodiments of the disclosed subject matter.

[0065] Note that, in some embodiments, a lid base **100** can include a drip-catch **208** (e.g., that catches liquid that comes out of an opening of lid base **100** when open), a lid base pocket **210**, a lid base rivet area **204**, and a rim **212**.

[0066] In some embodiments, drip-catch **208** of lid base **100** can have any suitable lid base pocket

210 with any type of depression, angle, edges, shape, arch, depth and width. In some embodiments, lid base pocket **210** can be used as a holding area for closure flap before lid base **100** is opened or when lid base **100** is re-opened. In some embodiments, lid base pocket **210** can be under a closure flap and can then be exposed once the closure flap is turned.

[0067] In some embodiments, drip-catch **208** can have a width or depth that is wide enough or deep enough to accommodate an opening of lid base **100** or a closure flap. As a more particular example, in an instance in which an opening of lid base **100** is wider than an opening of a traditional soda can, drip-catch **208** of lid base **100** can be relatively wider and more oval than a drip-catch of a traditional can end. In some embodiments, drip-catch **208** can include padding of any suitable type of material.

[0068] Referring to FIGS. **2A** and **2B**, shapes of rivet areas of a closure flap (e.g., closure flap **108** as shown in FIG. **2A**) and a lid base (e.g., lid base **100** as shown in FIG. **2B**) are shown in accordance with some embodiments of the disclosed subject matter. For example, referring to FIG. **2A**, in some embodiments, any suitable rivet can be inserted in a closure flap rivet area **202** to couple closure flap **108** to a tab (not shown). Note that, in some embodiments, a rivet inserted in closure flap rivet area **202** can be any suitable rivet, such as a rivet that is part of the tab (e.g., as shown in and described below in connection with FIG. **3E**), a rivet that is part of a lid base (e.g., as shown in and described below in connection with FIG. **10**), and/or an independent rivet (e.g., as shown in and described below in connection with FIGS. **4A**, **4B**, and **4C**). Additionally, note that, although closure flap rivet area **202** is shown as triangular in shape, in some embodiments, closure flap rivet area **202** can have any suitable shape (e.g., square, round, hexagonal, and/or any other suitable shape).

[0069] As another example, referring to FIG. **2B**, in some embodiments, any suitable rivet can be used in lid base rivet area **204** to connect lid base **100** to a tab and closure flap. Note that, in some embodiments, the rivet can be a rivet that is part of the lid base (e.g., as shown in and described below in connection with FIGS. **10A**, **10B**, and/or **10C**) that is located at lid base rivet area **204** on lid base **100**, a rivet that is part of the tab (e.g., as shown in and described below in connection with FIG. **3E**) that is inserted through the closure flap and through lid base rivet area **204**, and/or an independent rivet (e.g., as shown in and described below in connection with FIGS. **4A**, **4B**, and **4C**). Additionally, note that, although lid base rivet area **204** is shown as circular in shape, in some embodiments, lid base rivet area **204** can be any suitable shape (e.g., square, triangular, hexagonal, and/or any other suitable shape). In some embodiments, lid base rivet area **204** can be smaller or larger than closure flap rivet area **202**.

[0070] Note that, although tab **102**, closure flap **108**, and lid base **100** are shown in and described above in connection with FIGS. **1A**, **1B**, **2A**, and **2B**, in some embodiments, any suitable alternative designs of tabs, closure flaps, lid bases, and rivets can be used in any suitable combination. For example, in some embodiments, alternative tab designs (e.g., as shown in and described below in connection with FIGS. **3A**, **3B**, **3C**, **3D**, **3E**, **9A**, and/or **9F**), alternative closure flap designs (e.g., as shown in and described below in connection with FIGS. **6A**, **6B**, **6C**, **7A**, **7B**, **7C**, **8**, **9A**, **9B**, **9C**, **9D**, and/or **11**), alternative rivet designs (e.g., as shown in and described below in connection with FIGS. **4A**, **4B**, **4C**, **9A**, **9B**, **9C**, **9D**, **9E**, **9F**, **9G**, **10A**, **10B**, and/or **10C**), and/or alternative lid base designs (e.g., as shown in and described below in connection with FIGS. **10A**, **10B**, **10C**, and/or **11**) can be used in any suitable combination.

[0071] Turning to FIG. **3A**, a top view of tab **102** is shown in accordance with some embodiments of the disclosed subject matter. Note that tab **102** is shown in FIG. **3A** with a tab rivet area **301** that is triangular in shape. In some embodiments, a tab can have a rivet area with any suitable shape (e.g., circular, triangular, square, L-shaped, hexagonal, and/or any other suitable shape). For example, as shown in FIG. **3C**, a tab can have a round tab rivet area **312**.

[0072] Turning to FIG. **3B**, an angled side view of a tab **102** is shown in accordance with some embodiments of the disclosed subject matter. In some embodiments, a force part **302** of tab **102**

(e.g., a back portion of a tab) can have an arched or concave shape that allows tab **102** to be pulled upward by a finger and to be rotated. Additionally, note that tab **102** can have any suitable shape or dimensions that differ from tabs used in other drink cans. For example, in some embodiments, tab **102** can have a thicker force part **302** and/or thicker side portions **304** and/or **306** relative to tabs used in other drink cans. As another example, in some embodiments, a length and/or a width of tab **102** from force part **302** to tab rivet area **301** can be different in shape than tabs used in other drink cans (e.g., longer, shorter, wider, thinner, and/or different in any other suitable manner).

[0073] In some embodiments, a tab, a closure flap, and a lid base can be coupled using an independent rivet (e.g., a rivet that is not a part of any of the tab, the closure flap, and/or the lid base). FIGS. **4A**, **4B**, and **4C** show examples of independent rivets that can be used to couple any of a tab, a closure flap, and/or a lid base.

[0074] Turning to FIG. **4A**, example views of an independent rivet with a circular top portion are shown in accordance with some embodiments of the disclosed subject matter. View **400** shows a three-dimensional side view of a rivet. As illustrated, the rivet includes a circular top portion **402**, a triangular middle portion **404**, and a circular bottom portion **406**. In some embodiments, circular top portion **402** can overlap a tab and can lock into a tab rivet area and a closure flap rivet area, thereby holding both the tab and the closure flap to a lid base. In some embodiments, triangular middle portion **404** can act as a lock between the closure flap and the tab, thereby allowing the tab and the closure flap to rotate as one piece. In some embodiments, circular bottom portion **406** can go through lid base rivet area **204** (as shown in and described above in connection with FIG. **2B**) of lid base **100** thereby allowing the tab and the closure flap to rotate together. In some embodiments, circular bottom portion **406** can be flattened by machinery to secure the rivet, as shown in and described below in connection with FIG. **9G**.

[0075] FIG. **4A** additionally shows a top view **408** and a bottom view **410** of the rivet shown in view **400**. Note that, as shown in FIG. **4A**, in some embodiments, circular top portion **402** can have the largest area of the three portions of the rivet, and circular bottom portion **406** can have the smallest area of the three portions of the rivet. Note that, in some embodiments, each section of a rivet can vary in height, width, length, circumference, and/or in any other suitable manner.

[0076] Turning to FIG. **4B**, an example **412** of a rivet with a triangular top portion **414** is shown in accordance with some embodiments of the disclosed subject matter.

[0077] Turning to FIG. **4C**, an example **416** of a rivet with a round top and a cylindrical body **418** is shown in accordance with some embodiments of the disclosed subject matter.

[0078] In some embodiments, shapes of holes of a closure flap can correspond to portions of a rivet. For example, in some embodiments, rivet **416** (as shown in and described above in connection FIG. **4C**) can connect a tab with a circular tab rivet area to a closure flap with a circular closure flap rivet area. In some embodiments, this can allow the rivet and the closure flap to turn either clockwise or counter-clockwise.

[0079] It should be noted that the illustrated relative dimensions may differ from an actual rivet implemented in accordance with some embodiments. For example, in some embodiments, the height of each portion of **402**, **404**, **406**, and **418** may be shorter than a width and/or a diameter of that portion. In some embodiments, the heights may be exaggerated herein for clarity.

[0080] Turning to FIG. **5A**, an example top view of a scored flap of a lid base (e.g., such as scored flap **104** as shown in and described above in connection with FIG. **1B**) is shown in accordance with some embodiments of the disclosed subject matter. FIG. **5B** shows an example bottom view of the scored flap with perforations on the lid base. FIG. **5C** shows an example top view of an alternate curved up scored area **502** where it meets link **112**. FIG. **5D** shows an example top view of an alternate curved down scored area **504** where it meets link **112**. Note that, in some embodiments, any suitable scored flap can be used in connection with any lid base.

[0081] In some embodiments, a closure flap can include any suitable elements that can allow a tab to exert pressure on a closure flap when a closure flap is inserted into an opening of a can to allow

a more unified and/or tighter seal to be formed between the closure flap and the opening of the can. Additionally, a closure flap can include any suitable elements that allow a tab to rotate a connected closure flap. FIGS. **6A**, **6B**, and **6C** show examples of closure flaps with various design elements. Note that, in some embodiments, any of the elements shown in FIGS. **6A**, **6B**, and/or **6C** can be combined in any suitable manner in a closure flap.

[0082] As shown in FIGS. **6A**, **6B**, and **6C**, in some embodiments, a closure flap can have one or more variously shaped bumps, molds, raises, flaps, lips, or other designs. In some embodiments, a bump on a closure flap can aid a lock between a tab and the closure flap, aid in pushing and pulling the connected tab and closure flap, allow a tab to exert even pressure on a closure flap when the closure flap is inserted into an opening of a lid base, and/or perform any other suitable functions. In some embodiments, a closure flap can include any suitable bumps, such as bump **602** and/or bump **604** as shown in FIG. **6A**, bump **610** and/or bump **614** as shown in FIGS. **6A** and **6C**, and/or any other suitable bumps. In some embodiments, the one or more bumps can correspond to raised portions of an interior of the closure flap. Note, that, in some embodiments, a closure flap can include any suitable number of bumps (e.g., zero, one, two, three, five, and/or any other suitable number), each located at any suitable position on the closure flap. In some embodiments, the one or more bumps on a closure flap can be generated in any suitable manner, for example, by using any suitable welding or molding technique (e.g., using injection molding and/or any other suitable molding, welding, or shaping technique).

[0083] In some embodiments, the one or more bumps can meet a bottom portion of a tab, which can cause the tab to exert even pressure on the closure flap when the closure flap is inserted into an opening of an open can, thereby allowing a more unified and/or tighter seal to be formed between the closure flap and an opening of the can. In addition, the one or more bumps can act and engage as a lock with a tab to assist in rotation of the closure flap over an opening of the can end. Note that examples of closure flaps with one or more bumps that meet a bottom portion of a tab are shown in and described below in more detail in connection with FIGS. **7A**, **7B**, and **7C**. Turning to FIG. **3C**, in some embodiments, a square shaped bump (e.g., bump **614** of FIGS. **6A** and **6C**) can fit into a square shaped tab opening **314** of a tab **310**. As another example, in some embodiments, a round shaped bump (e.g., bump **610** of FIG. **6A**) can fit into a round shaped tab opening **318** of tab **310**. Note that, in some embodiments, each bump on a closure flap can have any suitable shape (e.g., round, square, rectangular, L-shaped, U-shaped, triangular, hexagonal, and/or any other suitable shape). Additionally, note that, in some embodiments, a tab can have openings of any suitable shape(s) that correspond to bumps of a closure flap that is to be connected to the tab.

[0084] Referring back to FIG. **6A**, top views of an alternate closure flap **114** are shown in accordance with some embodiments of the disclosed subject matter. In some embodiments, closure flap **114** can include one or more closure flap lips **612** (as shown in FIGS. **6A** and **6C**). In some embodiments, closure flap lips **612** can act as a lock or a hook that engages with lips on a tab. For example, turning to FIGS. **3C** and **3D**, tab **310** can include tab lips **316**. In some embodiments, as a tab turns, closure flap lips **612** can act as pressure points against tab lips **316**. In some embodiments, when the tab turns, both closure flap lips **612** and tab lips **316** can aid in the rotation (e.g., pushing or pulling a connected tab and closure flap), clockwise or counter-clockwise, and locking of the tab to closure flap **114** while still allowing the tab to rotate. Note that, referring to FIG. **3D**, in some embodiments, a tab can additionally include pressure points **320**. In some embodiments, when pressure is applied to pressure points **320**, the tab can engage with a connected closure flap to create a secure seal of the closure flap in an opening of a lid base. In some embodiments, pressure points **320** can be any suitable added material, bumps, and/or punctures. In some embodiments, pressure points **320** can be formed from any part of a mold and/or in any other suitable manner. Note that FIG. **3D** shows an underside of tab **310**.

[0085] Referring back to FIG. **6A**, in some embodiments, a closure flap can have an outer edge **609** that acts as an edge, lip, glide, flap, ledge, or seal. In some embodiments, outer edge **609** can

extend beyond opening **106** and can allow for a seal when the closure flap is positioned over and seals opening **106**. In some embodiments, outer edge **609** can additionally aid in rotation of the closure flap over the lid base.

[0086] Turning to FIG. **6B**, an example cross-sectional side view of a closure flap is shown in accordance with some embodiments of the disclosed subject matter. Note that, in some embodiments, side edges of a closure flap can have any suitable shape(s) (e.g., angles, curves, bends, arches, cuts, bumps, molds, and/or any other suitable designs), such as shown in and described below in more detail in connection with FIGS. **7A**, **7B**, **7C**, and **8**.

[0087] Turning to FIG. **6C**, an example side view of a closure flap is shown in accordance with some embodiments of the disclosed subject matter. Note that, in some embodiments, side edges of a closure flap can have any suitable shape(s), as shown in and described below in more detail in connection with FIGS. **7A**, **7B**, **7C**, **8**, **9B**, **9C**, and/or **9D**.

[0088] Turning to FIGS. **7A**, **7B**, and **7C**, example cross-sectional side views of closure flaps, with a tab and a lid base, of different shapes inserted into an opening of a lid base are shown in accordance with some embodiments of the disclosed subject matter. Note that FIGS. **7A**, **7B**, and **7C** each show three different potential edge shapes **702**, **704**, and **706**, respectively, of closure flaps. Additionally, note that the shapes of closure flaps shown in FIGS. **7A**, **7B**, and **7C** are shown merely as examples, and, in some embodiments, any suitable shape can be used. Note that additional possible shapes of closure flaps are shown in and described below in connection with FIGS. **8**, **9B**, **9C**, and/or **9D**. Additionally, note that, in FIGS. **7A**, **7B**, and **7C**, outer edge **609** of each closure flap is shown on top of lid base **100**.

[0089] Note that, in some embodiments, the closure flap and/or bumps on the closure flap can each be made of any suitable material. For example, in some embodiments, the closure flap and/or the bumps can each be made of the same material as the can, such as aluminum or an aluminum alloy. In some embodiments, the closure flap and/or the bumps on the closure flap can each be made of any suitable material that allows the closure flap and/or the bumps to be recycled along with the can.

[0090] Additionally, note that, in some embodiments, the closure flap can have a depressed pocket (e.g., a depressed pocket **708** as shown in FIG. **7A**) in the center that has dimensions that correspond to dimensions of an opening of an open can (e.g., opening **106** as shown in and described above in connection with FIG. **1B**).

[0091] In some embodiments, a bottom portion of the closure flap can have a lining sealant (e.g., lining sealant **710** as shown in FIG. **7A**) of any suitable material, such as the same coating as the interior of the can body or any other suitable material. Note that, in some embodiments, any suitable type of sealant can be used, such as a sealant used in a lining or interior of the can (e.g., epoxy resin, any suitable polymer plastic, and/or any other suitable sealant). In some embodiments, the sealant can be any suitable sealant that satisfies any suitable criteria, such as being harmless to a consumer (e.g., BPA-free, and/or any other suitable harmless sealant), recyclable with aluminum, does not contaminate and/or change a taste of a beverage in a can, and/or that satisfies any other suitable criteria. In some embodiments, the lining sealant can have any suitable layers and/or can have any suitable thickness that can aid in sealing the opening of the can end and/or can aid in ease of use. In some embodiments, the lining sealant can have any suitable dimensions to correspond to a portion of a bottom of a corresponding closure flap or to the entire bottom of a corresponding closure flap. Additionally or alternatively, note that, in some embodiments, the closure flap can be covered prior to first use with a breakable sealant that prevents contaminants from entering an area under the closure flap.

[0092] Turning to FIG. **8**, example cross-sectional side views of closure flaps of different shapes are shown in accordance with some embodiments of the disclosed subject matter. As illustrated, eight shapes, **802**, **804**, **806**, **808**, **810**, **812**, **814**, and **816** are shown with multiple design options and/or combination options. Note that, as illustrated in FIG. **8**, a closure flap can have any suitable

dimensions, and any suitable shape that includes any suitable angles, cuts, edges, padding, grooves, molds, lips, pressure points, curls, and/or any other suitable designs.

[0093] In some embodiments, a closure flap can have sections that can include: 1.) depressed pocket **708** (as shown in and described above in connection with FIG. 7A); 2.) a closure seal lock area **826**; and 3.) outer edge **609** (as shown in and described above in connection with FIG. 6). For example, in some embodiments, depressed pocket **708** can be various depths, angles, shapes, and/or widths in relationship to a closure seal lock area **826** for insertion inside of a lid base opening and/or insertion inside of a holding area of a lid base pocket (e.g., lid base pocket **210** of FIG. 2B), to accommodate movement of the closure flap (e.g., during rotation or positioning of the closure flap during opening or closing) and/or to aid in insertion of the closure flap in a lid base opening.

[0094] As another example, in some embodiments, as shown in shapes **812**, **814**, and **816**, the closure flap can include a straight edge **818** that is connected to angle drop **819** of any suitable angle (e.g., 40 degrees, 45 degrees, 50 degrees, and/or any other suitable angle), that leads to an under-cut/indent **820** that acts as a hook. In some embodiments, under-cut/indent **820** can be positioned underneath an edge of opening **106** once lid base **100** is open. In some embodiments, this can allow for a seal of the closure flap over opening **106** such that the seal does not interfere with link **112** but allows the closure flap to work in conjunction with link **112** to seal a lid base opening.

[0095] As yet another example, in some embodiments, outer edge **609** of shape **814** can act with a spring-like action when the closure flap is depressed in the middle via an attached tab to provide resistance or assistance when the closure flap is depressed in a lid base opening and/or when the closure flap is rotated by a coupled tab. Note that shape **814** includes an upward arch and downward bend **822** into a curled edge **824** that can tuck under the arch **822** and which can sit on top of a drip catch of a lid base, which can be used to provide the spring-like action, resistance, pressure, and/or any other suitable function to assist a tight and non-leakable seal of the closure flap over a lid base opening and a link of a lid base opening (e.g., link **112**, as shown in and described above in connection with FIG. 1B). Additionally, in some embodiments, curled edge **824** can provide a smooth surface area to an outer edge **609** of the closure flap such that the closure flap does not have an abrasive edge.

[0096] In some embodiments, a lid base can have a slide area in which a closure flap can move during rotation of the closure flap in and out of a lid base opening. In some embodiments, the slide area can aid in movement and/or rotation of the closure flap. Turning to FIG. 11, a top view of a lid base is shown in accordance with some embodiments of the disclosed subject matter. As illustrated, the lid base has a slide area **1102**. In some embodiments, slide area **1102** can be level with a scored flap of the lid base (e.g., scored flap **104**). Alternatively, in some embodiments, slide area **1102** can slope in any suitable direction (e.g., upward toward scored flap **104**, downward toward scored flap **104**, and/or in any suitable direction). Note that, in some embodiments, slide area **1102** can be any suitable portion of the lid base. For example, in some embodiments, as shown in FIG. 11, slide area **1102** can be a subset of the top area of the lid base (e.g., a right side of the lid base, a left side of the lid base, and/or any other suitable area of the lid base). As another example, in some embodiments, slide area **1102** can be a full top portion of the lid base. Note that, in some embodiments, slide area **1102** can be used in connection with any combination of any suitable tab, rivet(s), closure flap, scored area, link(s), and/or any other suitable elements of a can end described herein.

[0097] In some embodiments, as shown in FIG. 11, the lid base can include a push button **1103**. In some embodiments, push button **1103** can act as a pressure point. In some embodiments, push button **1103** can be depressed to aid in rotation of closure flap out and away from the lid base opening to re-open the can end by loosening a connection of the closure flap within the lid base opening. Note that, in some embodiments, push button **1103** can be included in any suitable lid base and can be located at any suitable position on the lid base. Additionally, note that, in some

embodiments, push button **1103** can be of any suitable shape and/or size (e.g., round, square, rectangular, triangular, L-shaped, and/or any other suitable shape). In some embodiments, push button **1103** can be formed via any suitable manufacturing technique(s) (e.g., any suitable molding technique, a pulled or pushed force manufacturing technique(s) as part of the lid base, welded onto the lid base, extruded from the lid base, and/or any other suitable technique(s)). In some embodiments, push button **1103** can be omitted.

[0098] Turning to FIGS. **9A**, **9B**, **9C**, **9D**, and **9F**, another example of a tab and a closure flap are shown in accordance with some embodiments of the disclosed subject matter.

[0099] Turning to FIG. **9A**, a top view of a tab **902** connected to a closure flap **906** and lid base is shown in accordance with some embodiments of the disclosed subject matter. As illustrated, tab **902** can include a tab rivet area **910**, a lip **916**, and/or a middle bar **918**. Note that FIG. **9F** shows a top view of tab **902**.

[0100] In some embodiments, tab rivet area **910** can correspond to a rivet that is used to couple tab **902** to the lid base. Note that, similar to what is described above, in some embodiments, a rivet used in connection with tab rivet area **910** can be any suitable rivet, such as an independent rivet (e.g., as shown in and described above in connection with FIGS. **4A**, **4B**, and/or **4C**), a rivet that is part of a lid base (e.g., as shown in and described below in connection with FIG. **10**), and/or a rivet that is part of tab **902** at a location of tab rivet area **910** (e.g., as shown in and described below in connection with FIG. **3E**).

[0101] In some embodiments, middle bar **918** can engage with closure flap **906** in any suitable manner. For example, in some embodiments, middle bar **918** can fasten to or otherwise engage with a platform area **908** (e.g., as shown in FIGS. **9B** and **9C**) of closure flap **906**. As another example, in some embodiments, tab opening **318** (as shown in FIG. **3C**) can fasten to and/or engage with a closure flap bump **904**, as shown in FIGS. **9A** and **9B**. As yet another example, in some embodiments, middle bar **918** can engage with any suitable independent rivet, such as a rivet **912** shown in FIG. **9E**.

[0102] Turning to FIG. **9B**, a side view of tab **902** and closure flap **906** is shown in accordance with some embodiments of the disclosed subject matter.

[0103] Turning to FIGS. **9C** and **9D**, a top view of closure flap **906** and a side view of closure flap **906**, respectively, are shown in accordance with some embodiments of the disclosed subject matter. As illustrated, closure flap **906** includes a bump **904** on platform area **908**. In some embodiments, bump **904** can engage with tab **902** by being inserted through a circular tab opening of tab **902** (e.g., tab opening area **318**, as shown in FIG. **9F**). Referring back to FIG. **9A**, in some embodiments, when bump **904** is inserted into the circular tab opening **318** of tab **902**, bump **904** can act as a rivet that couples tab **902** and closure flap **906**. For example, in some embodiments, bump **904** can be flattened (e.g., by a machine that applies pressure to bump **904**), thereby causing bump **904** to act as a rivet that couples tab **902** and closure flap **906**. Note that, in instances in which bump **904** acts as a rivet that couples tab **902** and closure flap **906**, bump **904** can couple tab **902** and closure flap **906** such that tab **902** and closure flap **906** rotate together in conjunction, thereby acting as a lock mechanism that can aid in tab **902** rotating closure flap **906**. In some embodiments, when bump **904** is flattened, bump **904** can create a lip (e.g., as shown in and described below in connection with FIG. **9G**) to keep tab **902** connected to closure flap **906**.

[0104] Note that, as shown in section **920** of FIG. **9C**, in some embodiments, platform area **908** can extend or protrude from an outer edge of the closure flap. In some embodiments, section **920** can increase a distance between rivet area **616** and a depressed pocket area of closure flap **906**. In some embodiments, platform **908** can have any suitable dimensions (e.g., length, width, thickness, etc.), and platform **908** can protrude from the outer edge of the closure flap by

[0105] Turning to FIGS. **10A**, **10B**, and **10C**, examples of a lid base **1002** with different lid base rivet area designs are shown in accordance with some embodiments of the disclosed subject matter. Note that, in some embodiments, lid base **1002** can have any suitable outer rim (e.g., an industry-

standard can end outer rim, and/or any other suitable outer rim), which can be attached and sealed to a body of a can in any suitable manner (e.g., a double-seam method, and/or in any other suitable manner). Note that, in some embodiments, any suitable tab and/or closure flap can be used in connection with any of the lid bases shown in FIGS. **10A**, **10B**, and/or **10C**.

[0106] Referring to FIG. **10A**, lid base **1002** can include a protuberance **1004**. In some embodiments, protuberance **1004** can be formed on lid base **1002** in any suitable manner. For example, in some embodiments, protuberance **1004** can be formed via any suitable manufacturing technique(s) (e.g., any suitable molding technique, a pulled or pushed force manufacturing technique(s) as part of lid base **1002**, welded onto lid base **1002**, extruded from lid base **1002**, and/or any other suitable technique(s)). In some embodiments, protuberance **1004** can be closed at a top portion of protuberance **1004**. In some embodiments, protuberance **1004** can function as a rivet that can couple a closure flap and a tab to lid base **1002**. For example, in some embodiments, protuberance **1004** can be inserted through a round shaped closure flap rivet area and through a round shaped tab rivet area. Continuing with this example, in some embodiments, protuberance **1004** can then be flattened in any suitable manner to allow protuberance **1004** to function as a rivet. In some embodiments, protuberance **1004** can be flattened in any suitable manner, such as by a machine that depresses protuberance **1004** and/or applies pressure to protuberance **1004** in any suitable manner.

[0107] Turning to FIG. **10B**, an example of lid base **1002** with a hollow protuberance **1006** is shown in accordance with some embodiments of the disclosed subject matter. In some embodiments, hollow protuberance **1006** can be formed in any suitable manner. For example, in some embodiments, hollow protuberance **1006** can be formed via any suitable manufacturing technique(s) (e.g., any suitable molding technique, a pulled or pushed force manufacturing technique(s) as part of lid base **1002**, welded onto lid base **1002**, extruded from lid base **1002**, and/or any other suitable technique(s)). In some embodiments, a hollow portion of hollow protuberance **1006** can be formed in any suitable manner. For example, in some embodiments, an adhered protuberance can be drilled, bored, extruded, and/or circular hole punched. In some embodiments, hollow protuberance **1006** can act as a column that can receive any suitable rivet used to couple a tab and a closure flap to lid base **1002**. For example, in some embodiments, hollow protuberance **1006** can receive an independent rivet, as shown in and described above in connection with FIGS. **4A**, **4B**, and/or **4C**. As another example, in some embodiments, hollow protuberance **1006** can receive a rivet that is part of a tab, as shown in and described below in connection with FIG. **3E**. In some embodiments, a rivet inserted in hollow protuberance **1006** can be sealed or attached from an underside of lid base **1002** in any suitable manner (e.g., as shown in and described below in connection with FIG. **3F**). For example, in some embodiments, the rivet can be flattened in any suitable manner, such as by a machine that depresses the rivet and/or applies pressure to the rivet in any suitable manner.

[0108] Turning to FIG. **10C**, an example of a lid base **1002** with a rivet opening **1008** is shown in accordance with some embodiments of the disclosed subject matter. In some embodiments, rivet opening **1008** can be formed in any suitable manner. For example, in some embodiments, rivet opening **1008** can be formed from any suitable mold, can be circular hole punched in lid base **1002**, bored in lid base **1002**, drilled through lid base **1002**, and/or formed in any other suitable manner. In some embodiments, any suitable rivet can be inserted through rivet opening **1008** to couple a tab and a closure flap to lid base **1002**. For example, in some embodiments, rivet opening **1008** can receive an independent rivet, as shown in and described above in connection with FIGS. **4A**, **4B**, and/or **4C**. As another example, in some embodiments, rivet opening **1008** can receive a rivet that is part of a tab, as shown in and described below in connection with FIG. **3E**.

[0109] Turning to FIG. **3E**, examples of rivets that are part of a tab are shown in accordance with some embodiments of the disclosed subject matter. As illustrated, FIG. **3E** shows different designs for rivets that can be attached to or are otherwise part of a tab **350**. Note that, in some

embodiments, tab **350** can be used in connection with any suitable closure flap and/or any suitable lid base that can receive a rivet that is part of tab **350**. Additionally, note that, in some embodiments, a rivet that is part of tab **350** can be formed in any suitable manner. For example, in some embodiments, a rivet can be adhered to tab **350** (e.g., welded to tab **350**, pulled or pushed force or other suitable manufacturing technique as part of tab **350**, and/or adhered in any other suitable manner), formed via any suitable mold, extruded, and/or formed in any other suitable manner.

[0110] In some embodiments, a rivet that is a part of tab **350** can be any suitable shape, and can include any suitable sections, each having any suitable shape. For example, in some embodiments, tab **350** can include a rivet **351** that has a circular shape. As another example, in some embodiments, tab **350** can include a rivet **352** that can include a triangular section that can act as a lock above a circular portion of rivet **352**. As yet another example, in some embodiments, tab **350** can include a rivet **353** that includes a square section that can act as a lock above a circular portion of rivet **353**.

[0111] In some embodiments, a circular portion of a rivet that is part of tab **350** can be inserted through a circular closure flap rivet area (e.g., circular closure flap rivet area **616**, as shown in FIG. **6A**, and/or any other suitable circular closure flap rivet area) and then through a circular lid base rivet area (e.g., lid base rivet area **204** as shown in FIG. **2B**, hollow protuberance **1006** as shown in FIG. **10B**, rivet opening **1008** as shown in FIG. **10C**, and/or any other suitable circular lid base rivet area).

[0112] In some embodiments, a rivet can then be attached to the connected lid base, closure flap, and tab in any suitable manner. For example, in some embodiments, the rivet can be flattened in any suitable manner, such as by a machine that depresses the rivet from an underside of the lid base after the rivet is inserted through the lid base rivet area. Turning to FIG. **3F**, an underside of a lid base after a rivet is inserted and flattened is shown in accordance with some embodiments of the disclosed subject matter. As illustrated, FIG. **3F** includes a flattened rivet **354** that has been flattened by any suitable machine that applies pressure to the rivet after insertion through the lid base rivet area.

[0113] Turning to FIG. **9G**, an example cross-sectional view of a machine flattened rivet is shown in accordance with some embodiments of the disclosed subject matter. As illustrated, FIG. **9G** includes a flattened rivet **930**, and a side portion **932**. In some embodiments, flattened rivet **930** can correspond to any suitable rivet that is created using any suitable technique(s), as described above. Note that, in some embodiments, flattened rivet **930** can correspond to an independent rivet, a rivet that is part of a tab, a rivet that is part of a lid base, and/or any other suitable rivet. For example, in some embodiments, flattened rivet **930** can correspond to an independent rivet as shown in and described above in connection with FIGS. **4A**, **4B**, and/or **4C**. As another example, in some embodiments, flattened rivet **930** can correspond to a rivet that is part of a tab as shown in and described above in connection with FIG. **3E** and/or a rivet that is part of a lid base as shown in and described above in connection with FIG. **10A** or **10B**.

[0114] In some embodiments, side portion **932** can correspond to any suitable surface on which a lip of flattened rivet **930**, once flattened, is positioned. For example, in an instance in which flattened rivet **930** corresponds to a rivet that is part of a tab or an independent rivet that is inserted through a tab, closure flap, and lid base, side portion **932** can correspond to a lid base, where flattened rivet **930** is flattened on an underside of the lid base. As another example, in an instance in which flattened rivet **930** corresponds to a rivet that is part of a lid base, side portion **932** can correspond to a tab, where flattened rivet **930** is flattened from above a tab.

[0115] Turning to FIGS. **12-14**, an example **1200** of a can end assembly in accordance with some embodiments is shown. As illustrated, can end assembly **1200** include a can end **1202**, a closure **1204**, a tab **1206**, and a center rivet **1208**, in some embodiments.

[0116] Can end **1202**, closure **1204**, tab **1206**, and center rivet **1208** can be made from any suitable

material in some embodiments. For example, in some embodiments, can end **1202**, closure **1204**, tab **1206**, and/or center rivet **1208** can be made from aluminum, tin, stainless steel, or any suitable combination of the same.

[0117] In some embodiments, can end **1202**, closure **1204**, and center rivet **1208** can include any suitable coating on an interior or exterior surface of the can end, the closure, and the center rivet. For example, in some embodiments, can end **1202**, closure **1204**, and/or center rivet **1208** can include an epoxy resin, BPA, acrylic, polyester, PVC from vinyl, oleoresins, or other chemicals coating on the interior surface of the can end.

[0118] In some embodiments, can end **1202** can be formed in any suitable manner. For example, in some embodiments, can end **1202** can be formed by stamping, cutting, molding, or laser cutting the can end out of sheet material, such as sheet aluminum.

[0119] In some embodiments, closure **1204** can be formed in any suitable manner. For example, in some embodiments, closure **1204** can be formed by stamping, cutting, molding, or laser cutting the closure out of sheet material, such as sheet aluminum.

[0120] In some embodiments, tab **1206** can be formed in any suitable manner. For example, in some embodiments, tab **1206** can be formed by stamping, cutting, molding, or laser cutting the tab out of sheet material, such as sheet aluminum.

[0121] In some embodiments, center rivet **1208** can be formed in any suitable manner. For example, in some embodiments, center rivet **1208** can be formed by extruding, pressing, stamping, or molding the center rivet from rod material, such as rod aluminum.

[0122] Further features of a can end in accordance with some embodiments are described below in connection with FIGS. **15** and **16**. Further features of a closure in accordance with some embodiments are described below in connection with FIGS. **17-21**. And, further features of a tab in accordance with some embodiments are described below in connection with FIGS. **22-24**.

[0123] Turning to FIGS. **15** and **16**, a can end **1502** in accordance with some embodiments is shown. Can end **1502** can be used as can end **1200** in some embodiments.

[0124] As illustrated, can end **1502** can include a recess portion **1504**, a tear strip **1506**, and a tear line **1508**. When can end **1502** is unopened, tear strip **1506** is in a sealed position shown in FIG. **15** and tear strip, tear line, and can end are liquid, air, and gas tight. When a tab, such as tab **1206** as pressed with suitable force against the tear strip, the tear line breaks and the tear strip bends downward into an opened position.

[0125] In some embodiments, recess **1504** can have any suitable shape and dimensions. For example, in some embodiments, recess **1504** can have a deepest portion **1510** that has a shape and dimensions that generally matches the shape of the underside of a portion of a closure, such as closure **1204**, when the closure is located therein. As another example, in some embodiments, recess **1504** can have a shallowest portion **1510** that is at a same height relative to a top portion **1512** of the can end as tear strip **1506** is relative to the top portion when tear strip **1506** is in the sealed position. Such a configuration can allow the closure to slide up out of recess **1504** and deepest portion **1510** from the deepest portion to the shallowest portion and around to the tear strip and close the hole that is present when the tear strip is in the opened position.

[0126] Turning to FIGS. **17** and **18**, an example **1702** of a closure that can be used as closure **1204** in accordance with some embodiments is shown. As illustrated in FIG. **17**, closure **1702** is on a can end without a tab, such as tab **1206**, on top of the closure. In this figure, the closure is at the 12 o'clock position on the can end. In use, the closure can rotate clockwise from the 12 o'clock position to the 6 o'clock position on the can end.

[0127] Closure can have any suitable shape in some embodiments. For example, the shape of the closure can allow it to close the hole that is present when tear strip **1506** is in the opened position.

[0128] As shown in FIG. **18**, closure **1702** includes bumps **1704** and **1706**, closure rivet **1708**, a closure ear **1710**, a closure hole **1712**, and a closure ledge **1703** in some embodiments.

[0129] In some embodiments, bumps **1704** and **1706** can have any suitable size and shape. For

example, in some embodiments the bumps can be tall enough to touch the bottom of a tab, such as tab **1206**, when positioned on top of it. The bumps can have any suitable shape, such as round (as shown), oval, elliptical, square, triangular, etc.

[0130] Rivet **1708** can be positioned to engage a tab, such as tab **1206**, as shown in FIG. **12**, in some embodiments. The rivet can be separate from or part of the remainder of closure **1702**.

[0131] Closure ear **1710** can have any suitable size and shape in some embodiments. For example, the bottom portion of the ear can be rounded as shown in FIG. **18**, flat, or multi-sided.

[0132] Closure hole **1712** can have any suitable size, such as a size large enough to pass a center rivet, such as center rivet **1208**, and spin around the rivet without being too tight to easily turn with a person's fingers and without being too loose so as to not automatically align with a hole in the can end when the closure is at the 6 o'clock position relative to the can end as described above.

[0133] Closure ledge **1703** can have any suitable size and shape in some embodiments. For example, in some embodiments, closure ledge can overlap with an area surrounding the opening of the can end when in the opened position. Any suitable overlap can be used in some embodiments, such that liquid does not leak from the overlap, in some embodiments.

[0134] FIGS. **19-20** show an alternate example **1902** of a closure in accordance with some embodiments. As shown, closure **1902** includes bumps **1904** and **1906**, closure rivet **1908**, and closure ledge **1903** which can be the same as or similar to bumps **1704** and **1706**, closure rivet **1708**, and closure ledge **1703** as described above. As also shown, closure **1902** omits closure ear **1710** and closure hole **1712**.

[0135] As shown in FIG. **21**, the bottom of the closure (e.g., closures **1702** and/or **1902**) can include an insertion portion **2102** that enters the opening formed when a can end is in the opened position. The outer circumference of the insertion portion can be shaped to match and engage with the inner circumference of the opening in some embodiments. As also shown in FIG. **21**, the outer circumference of the insertion portion can have an angle α with respect to a vertical axis in the figure. In some embodiments, the angle α can be constant around the outer circumference of the insertion portion, while in other embodiments the angle α can vary over the course of different portions of the outer circumference of the insertion portion. In some embodiments, the angle α can smoothly vary over the outer circumference of the insertion portion.

[0136] FIGS. **22-24** show an example **2200** of a tab that can be used as tab **1206** in accordance with some embodiments. As illustrated, tab **2200** includes a rear side **2202**, a front side **2204**, a left side **2206**, a right side **2208**, and a center portion **2210** (outlined by the dashed line in FIG. **22**).

[0137] Each of rear side **2202**, front side **2204**, left side **2206**, right side **2208**, and center portion **2210** can have any suitable dimensions in some embodiments.

[0138] The center of the rear side **2202** can be bent inward (as shown) or outward, or be straight, in some embodiments. In some embodiments, the center of the rear side can additionally or alternatively be bent upward or downward, or be straight.

[0139] As shown, front side **2204** can be asymmetrical, in some embodiments. The forward-most point **2212** of the front side can be any where between the outermost edges of left side **2206** and right side **2208**, in some embodiments. The curve formed by the front of the front side can have any suitable shape, such as that shown in FIGS. **22** and **23**.

[0140] Left side **2206** and right side **2208** can have any suitable lengths, and be the same length or different lengths (as shown), in some embodiments. Left side **2206** and right side **2208** can each be straight (as shown), or be bent inward, outward, upward, and/or downward, in some embodiments.

[0141] In some embodiments, the tab can have any suitable bumps, depressions, indents, stamps, heights, ramps, cut-outs, or curling.

[0142] Center portion **2210** can have a center rivet hole **2214**, a bend **2216**, a downward bump **2218**, and a closure rivet hole **2220**, in some embodiments. The center rivet hole can have any suitable size and shape, in some embodiments. The bend can change the level of a rearward portion **2222** compared to a forward portion **2224** by any suitable amount. For example, as shown in FIG.

24, rearward portion **2222** is lower than the forward portion, which is not visible. The downward bump can have any suitable size and shape, in some embodiments. The closure rivet hole can have any suitable size and shape, in some embodiments.

[0143] Center portion **2210** can have any suitable size in some embodiments. For example, in some embodiments, can position center hole **2214** further from front portion **2204** than in other embodiments depending on the specific implementation. Any suitable distance between the forward-most portion of the center portion and the point **2212** can be used in some embodiments. Similarly, any suitable distances between each of center rivet hole **2214**, bend **2216**, downward bump **2218**, and closure rivet hole **2220** can be used in some embodiments, and these distances can be selected based on a particular application in some embodiments.

[0144] In accordance with some embodiments, a closure that can be used as closure **1204** can include a safety feature that can make the can end difficult to open. For example, in some embodiments, such a safety feature can be used to make it difficult for the can end from being opened by a child or a person with a handicap.

[0145] Turning to FIGS. **25-27**, an example of a closure incorporating a safety feature in accordance with some embodiments is illustrated. As shown, in some embodiments, the closure may include a main portion **2502**, that can be the same or similar to that described above for closure **1702**, and a bar portion **2504**. The bar portion can have any suitable shape and size, in some embodiments. For example, the bar portion can be substantially rectangular and be longer than the distance from top to bottom in the figure across tear strip **1506** and tear line **1508**, in some embodiments. In this way, if there is an attempt to open the can end by lifting the rear portion of the tab, the front portion of the tab hits the bar portion, rather than the tear strip, and cannot force the tear strip downward. In some embodiments, bar portion **2504** can include a raised portion **2506** that makes grabbing the bar portion with one's finger or fingernail easier. Raised portion **2506** can have any suitable size and shape, and be in any suitable location, in some embodiments. In some embodiments, bar portion **2504** can include a break-away portion **2508** that enables the bar portion to be removed from the remainder to the closure by a user. For example, with reference to the view in FIG. **25**, a user may grab the bar portion by the raised portion and rotate it counter-clock-wise. This may cause the bar portion to break away from the remainder of the closure at the break-away portion, so that the tab can then be used to open the can end. The break-away portion can include any suitable features to facilitate this process, such as scoring between the bar portion and the main portion of the closure, one or more cut-outs, thinner material, and/or any other suitable feature(s).

[0146] Turning to FIGS. **28-30**, another example of a closure incorporating a safety feature in accordance with some embodiments is illustrated. As shown, in some embodiments, the closure may include a main portion **2802**, that can be the same or similar to that described above for closure **1702**, and a bar portion **2804**. The bar portion can have any suitable shape and size, in some embodiments. For example, the bar portion can be partially triangular (as shown in FIG. **29**) and span across any suitable portions of the area surrounding tear strip **1506** and tear line **1508**. As another example, the bar portion can be flipped compared to what it is shown in FIGS. **28-30** so that it connects to the main portion on its right side rather than its left side. In this way, if a child attempts to open the can end by lifting the rear portion of the tab, the front portion of the tab hits some part of the bar portion, rather than the tear strip, and cannot force the tear strip downward. In some embodiments, bar portion **2804** can include a raised portion **2806**, that may or may not attach to **2802**, that makes grabbing the bar portion with one's finger or fingernail easier. Raised portion **2806** can have any suitable size and shape, and be in any suitable location, in some embodiments. In some embodiments, bar portion **2804** can include a break-away portion **2808** that enables the bar portion to be removed from the remainder to the closure by a user. For example, with reference to the view in FIG. **28**, a user may grab the bar portion by the raised portion and rotate it counter-clock-wise, or other directions. This may cause the bar portion to break away from the main portion of the closure at the break-away portion, so that the tab can then be used to open the can end. The

break-away portion can include any suitable features to facilitate this process, such as scoring between the bar portion and the main portion of the closure, one or more cut-outs, thinner material, and/or any other suitable feature(s).

[0147] Accordingly, reclosable can ends are provided.

[0148] Although the invention has been described and illustrated in the foregoing illustrative embodiments, it is understood that the present disclosure has been made only by way of example, and that numerous changes in the details of implementation of the invention can be made without departing from the spirit and scope of the invention, which is limited only by the claims that follow. Features of the disclosed embodiments can be combined and rearranged in various ways.

Claims

1. A reclosable can end, comprising: a lid base, wherein the lid base comprises a scored area; a tab, wherein a first end of the tab is configured to apply pressure to the scored area of the lid base when a second end of the tab, opposite the first end of the tab, is lifted and causes the scored area to at least partially separate from the lid base to expose an opening in the lid base; a closure flap, wherein the closure flap comprises: a first portion that has dimensions corresponding to dimensions of the opening in the lid base, wherein the first portion of the closure flap connects with a portion of the tab; a second portion that comprises an outer edge that extends beyond the first portion of the closure flap; a protruding portion that protrudes from the outer edge of the second portion of the closure flap and includes an opening; at least one bump that can contact the second end of the tab when tab is in a downward position; and a rivet that connects the tab and the closure flap to the lid base such that the tab and the closure flap are configured to rotate around the rivet, wherein the rivet connects the tab and the closure flap to the lid base via the opening in the protruding portion of the closure flap, wherein the first portion of the closure flap is configured to be inserted into the opening in the lid base when the tab and the closure flap are rotated to a position corresponding to a position of the opening in the lid base, and wherein the outer edge of the second portion of the closure flap is positioned on a top portion of the lid base when the first portion of the closure flap is inserted into the opening in the lid base.
2. The reclosable can end of claim 1, wherein the lid base comprises a lid base pocket that has dimensions that correspond to the dimensions of the first portion of the closure flap.
3. The reclosable can end of claim 2, wherein the lid base comprises a depressed slide area formed from the lid base pocket to the opening in the lid base.
4. The reclosable can end of claim 1, wherein a bottom portion of the first portion of the closure flap includes a coating of sealant.
5. The reclosable can end of claim 4, wherein the coating of sealant comprises a plurality of layers of sealant.
6. The reclosable can end of claim 4, wherein the coating of sealant comprises a material that is harmless to consumers.
7. The reclosable can end of claim 1, wherein the second end of the tab is concave such that the second end curves toward the first end.
8. The reclosable can end of claim 1, wherein the first portion of the closure flap includes a raised bump that connects the closure flap to the tab.
9. The reclosable can end of claim 1, wherein the outer edge of the second portion of the closure flap is flat.
10. The reclosable can end of claim 1, wherein the outer edge of the second portion of the closure flap is angled upward away from the first portion of the closure flap.
11. The reclosable can end of claim 1, wherein the outer edge of the second portion of the closure flap has an arched shape that bends away from the first portion of the closure flap.
12. The reclosable can end of claim 1, wherein the outer edge of the second portion of the closure

flap has an arched shape that bends towards the first portion of the closure flap.

13. The reclosable can end of claim 1, wherein the outer edge of the second portion of the closure flap has an arched shape.

14. The reclosable can end of claim 1, wherein the rivet is formed as part of the top portion of the lid base.

15. The reclosable can end of claim 1, wherein the rivet is formed as part of the tab.

16. The reclosable can end of claim 1, wherein the rivet is not formed from any of the lid base, the tab, and the closure flap.

17. The reclosable can end of claim 1, wherein the lid base comprises a drip catch.

18. A reclosable can end, comprising: a lid base, wherein the lid base comprises a scored area; a tab, wherein a first end of the tab is configured to apply pressure to the scored area of the lid base when a second end of the tab, opposite the first end of the tab, is lifted and causes the scored area to at least partially separate from the lid base to expose an opening in the lid base; a closure flap, wherein the closure flap comprises: a first portion that has dimensions corresponding to dimensions of the opening in the lid base, wherein the first portion of the closure flap connects with a portion of the tab; a second portion that comprises an outer edge that extends beyond the first portion of the closure flap; and a protruding portion that protrudes from the outer edge of the second portion of the closure flap and includes an opening; and a rivet that connects the tab and the closure flap to the lid base such that the tab and the closure flap are configured to rotate around the rivet, wherein the rivet connects the tab and the closure flap to the lid base via the opening in the protruding portion of the closure flap, wherein the first portion of the closure flap is configured to be inserted into the opening in the lid base when the tab and the closure flap are rotated to a position corresponding to a position of the opening in the lid base, wherein the outer edge of the second portion of the closure flap is positioned on the top portion of the lid base when the first portion of the closure flap is inserted into the opening in the lid base, and wherein the lid base comprises a push button that is positioned to serve as a stop against movement of the closure flap and the tab, but may be selectively depressed to permit movement of the closure flap and the tab.

19. The reclosable can end of claim 20, wherein the lid base comprises a lid base pocket that has dimensions that correspond to the dimensions of the first portion of the closure flap.

20. The reclosable can end of claim 19, wherein the lid base comprises a depressed slide area formed from the lid base pocket to the opening in the lid base.
